

Individualized Screening and In-Situ Health Monitoring for TID-Induced Effects in a Complex Microcontroller

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Abstract—Inter-device variability in TID-induced failure was observed across ninety-seven MSP430FR6989 microcontroller units spanning two fabrication lots, with a range in observed failure of roughly 120 krad(SiO₂). Programmable bias and inherent manufacturing variability have been shown to be dominant factors in determining this failure point. Predictive models informed by bias configurations and internal variability, in the form of lot membership and pre-irradiation characteristics of a device's internal oscillator, are presented, demonstrating a method for non-destructive screening of individual devices for beyond worst-case qualification of complex SoCs. During irradiation, characteristics of the device's local oscillators are also shown to be effective monitors of device health. The internal very-low-frequency oscillator (VLO) features enable prediction of the dose the part can receive before failure at statistical parity with an external dose record, while rail-to-rail voltage swings common to the oscillators in the output line sharpen that estimate as failure approaches.

Index Terms—Bias, Failure, Imminent, In-situ, MCU, Optimization, Oscillator, Pre-rad, Predict, Programming, TID, SoC

I. INTRODUCTION

MICROCONTROLLERS are often capable of entering a variety of sleep states, enabling power conservation during operation. Especially in power-constrained environments, the choice of sleep state can be leveraged to great impact. However, some of these constrained environments are accompanied by additional challenges, such as the presence of ionizing radiation. Because of these challenges, manufacturers and application designers are tasked with qualifying their parts to withstand specific radiation thresholds. During these qualifications, devices are typically tested in mostly static states, and the full range of programming strategies and sleep states is not investigated [1], [2].

In addition, process variability between devices can yield markedly different performance in harsh environments [3]. These variations often yield no significant effects in nominal operations. However, the effects of ionizing radiation, such as total ionizing dose (TID), can be exacerbated. TID occurs in semiconductor devices due to the presence of trapping centers

in bulk oxides and at the interfaces of the oxides with the semiconductor materials [4], [5]. Authors in [3] illustrate the varying effects between 16 nm bulk nFinFET devices, showing that transistors with baseline drain current higher than that of their counterparts exhibited the greatest degradation.

To measure the range of effects that process variability can induce, multiple devices are often tested, yielding a worst-case prediction [6]. These investigations provide valuable information about the part's baseline performance, but they can fall short for designers who want to maximize their application's lifetime. Because of the modern drive to use COTS devices in harsh environments, a need to gauge the inherent device-to-device variability that can lead to diverse effects has arisen. An improved understanding of failure patterns due to innate variability and programming decisions could allow designers to better estimate and potentially extend mission lifetimes. Such pre-irradiation screening would supplement conventional radiation lot-acceptance testing, extending an approach previously demonstrated for linear bipolar parts, where pre-irradiation electrical parameters were shown to track per-part post-irradiation response [7].

Here, variability in TID-induced functional failure ranging from 72 to 193 krad(SiO₂) is observed through testing of ninety-seven Texas Instruments (TI) MSP430FR6989 (MSP430) commercial off-the-shelf (COTS) microcontroller units (MCUs) drawn from two fabrication lots across two test campaigns. These MCUs include multiple internal bias conditions in the form of programmable low power modes (LPMs) that selectively turn off the CPU, internal oscillators, and peripherals. The choice of these LPMs has previously been shown to affect performance degradation with TID [8], [9].

In this study, the first observed point of failure is analyzed. Distributions of failure points between programmable bias states and fabrication lots are shown to be distinct, and knowledge of the bias state and lot membership pre-irradiation is shown to improve predictions regarding device lifetime. The baseline behavior of the internal very low-frequency oscillator (VLO) is shown to further improve these predictions in bias modes LPM0, LPM2, and LPM3, though end-to-end permutation testing shows those margins do not yet separate from the selection process at this sample size. Trained models are also shown to leverage features of the VLO's timing network to effectively separate the two fabrication lots, contributing to lot-blind predictive models of the most tolerant devices pre-irradiation.

During irradiation, VLO behavior is further shown to sup-

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port an in-situ prognostic capability, providing individualized dose-to-failure estimates without additional hardware. The continuous dose tracking throughout life proves specific to the VLO, while the end-of-life signal that flags imminent failure also appears on the factory-trimmed DCO captured on the same pin, demonstrating potential generalization to any observable output clock. Throughout analysis, the DCO serves as the on-chip control that isolates each signal to its source.

II. EXPERIMENTAL DESIGN

A. System Setup

One hundred seventy TI MSP430FR6989 devices were programmed independently to enter either a single LPM or a combination of LPMs during irradiation. Two test campaigns were conducted, drawing from two fabrication lots. Both campaigns occurred at ambient temperature in the same irradiation chamber, using identical test hardware and firmware. The principal differences were limited to the devices under test and the testing time. Only the ninety-seven devices that were statically biased are analyzed here (45 from the first lot, 52 from the second). Two first-lot devices lacking pre-irradiation baseline captures are excluded from all modeling, leaving ninety-five: 18 in active mode (AM), 16 in LPM0, 16 in LPM1, 14 in LPM2, 16 in LPM3, and 15 in LPM4. TABLE I details device operation in these states.

A custom board (Fig. 1, bottom left) sockets four devices under test (DUT)s for simultaneous irradiation. This board includes connections to and from the DUTs, including UART, power, I/O control signals, and the SMCLK output clock signal. MSP430FR6989 devices were pre-programmed via JTAG and placed on the test board in groups with identical bias conditions. The control signals are routed to another custom breakout board with an ESP32 microcontroller, shown in the upper-middle of the block diagram, which handles I/O control of the 4 DUTs, multiplexer control determining which DUT output was placed on the SMCLK line, oscilloscope (Keysight DSOX1204G) connections for clock-signal collection, and UART connections to the main control PC. The control PC is responsible for orchestrating experimental timing, the ESP32's control via UART commands, oscilloscope (o-scope) setup, data collection, and logging.

B. Irradiation and Procedure

All irradiations were performed in a temperature-controlled Co-60 chamber at dose rates compliant with MIL-STD-883, Test Method 1019.9, Condition A (standard condition). An initial capture (baseline) was taken prior to plunge into the irradiation chamber. An additional capture was taken immediately upon chamber entrance. Following this, the DUTs were placed into an LPM via an I/O handshaking protocol with the ESP32. The control PC then started an 80-second timer in which the DUTs were irradiated. The DUTs would then be woken briefly via an I/O handshaking protocol, during which the VLO and DCO clock signals would be output via the SMCLK line sequentially. The o-scope would be triggered by a signal from the ESP32 following a clock switch to capture the clock signal in 50 ms windows at 20 MHz. Finally, the

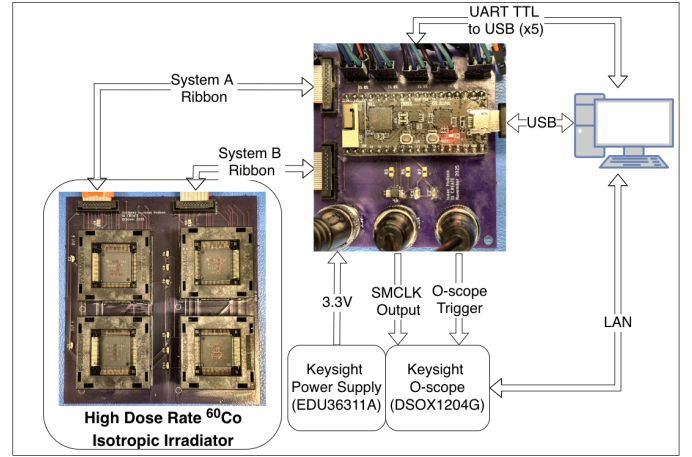


Fig. 1. Block diagram of the hardware setup used during irradiation. The bottom left block represents the 4x DUT board placed in the irradiation chamber, while the upper-middle board shows the ESP32 interface board.

MSP430 devices were programmed to capture their own VLO frequency (which typically ranges from 6 to 14 kHz) using an internal timer clock sampled by the DCO at 1 MHz, and output the average period of 100 captures via UART. This cycle would repeat until all 4 DUTs on the test board exhibited a functional failure, as defined by a lack of clock output on the SMCLK line, a failure to respond to I/O interrupts, or a failure to communicate via UART. Because interrogation occurred at discrete dose steps, the failure dose of each device is known only as an interval bounded by its last functional and first failed observations. Additionally, several devices survived the test limit and are treated as right-censored in the pre-irradiation models.

III. EXPERIMENTAL RESULTS

The cumulative distributions of the first functional failure, stratified by bias mode and manufacturing lot, are presented in Fig. 2. These results illustrate extreme variability in the observed point of first functional failure among the 97 statically biased DUTs. Both bias mode and fabrication lot drive the spread, with the programmed sleep state alone shifting typical failure doses by tens of krad(SiO₂) from the least restrictive modes to the most power-conservative. The lot effect is large, and its size depends strongly on the bias mode. Pooled over all modes, average failure doses differ by roughly 18 krad(SiO₂) between the two lots. In AM, second-lot devices failed at an average of 180 krad(SiO₂) against 98 for the first lot. Four of the eight second-lot AM devices were only tested to 150 krad(SiO₂) and were not observed to fail. These devices are excluded from the average. This limit was set before the campaign when survival to that point was not anticipated. LPM0 shows a comparable separation, with a second-lot average of 170 krad(SiO₂) against 98 for the first lot, whose AM and LPM0 averages happen to coincide. The remaining modes show little lot separation at all. LPM1 average failures between lots lie within 9 krad(SiO₂) of one another, the first lot being the more tolerant. The LPM2 through LPM4 lot averages agree within 2 krad(SiO₂), making the most power-conservative modes the most lot-consistent in the cohort. Within both lots, devices biased in power-conservative states

TABLE I
LOW-POWER MODE SUMMARY (DERIVED FROM TI DATASHEET INFORMATION [10])

Mode	AM	LPM0	LPM1	LPM2	LPM3	LPM4
CPU state	on	off	off	off	off	off
FRAM state	on	standby ^a	off	off	off	off
Max system clock	16 MHz	16 MHz	16 MHz	50 kHz	50 kHz	0
Typical current (25 °C)	103 μ A/MHz	75 μ A @ 1 MHz	40 μ A @ 1 MHz	0.9 μ A	0.4 μ A	0.3 μ A
MCLK	on	off	off	off	off	off
SMCLK	optional	optional	optional	off	off	off
ACLK	on	on	on	on	on	off
SVS	always	always	always	optional	optional	optional

^a In LPM0, FRAM is typically in standby; some configurations may fully power it off.

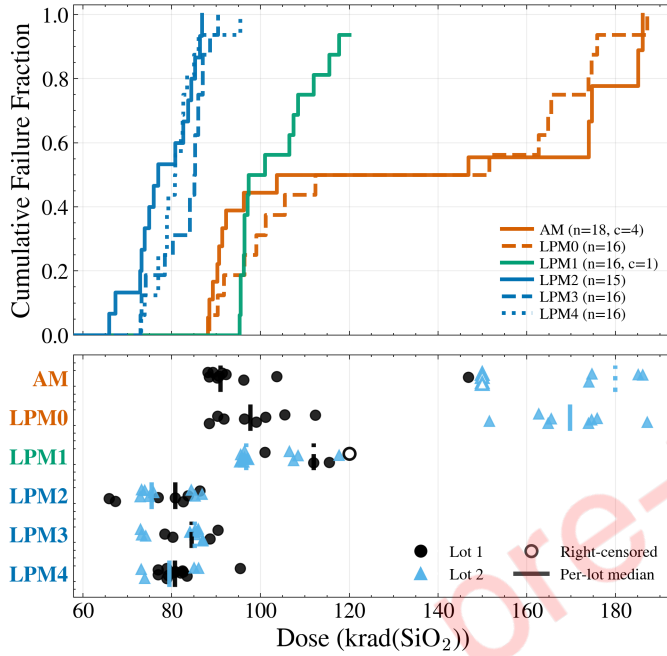


Fig. 2. Distributions of the first functional failure for the 97 statically biased devices, separated by bias mode and fabrication lot.

generally failed at the lowest doses. Even among identically biased devices from the same lot, however, major variations in the first point of failure are evident. No failure interval begins below roughly 60 krad(SiO₂), establishing a failure-free region for worst-case qualification.

The device behavior at and after the first functional failure point was not wholly consistent between devices. In both campaigns, devices first failed to respond to an I/O-driven interrupt tasked with either taking the device out of sleep or switching the clock output. For approximately half of the second-campaign devices, this error was paired with simultaneous loss of clock output and failure to communicate over UART. No device, however, lost its clock signal over the SMCLK line prior to failing to respond to I/O. The primary discrepancy in failure behavior is therefore the timing gap between the first digital failure, defined here by an unresponsive I/O interrupt, and complete clock failure. In the second campaign, 23 of the 48 failed devices exhibited coincident digital and analog failure at the first failure observation. Nine lost the clock one interrogation step, or approximately 12 krad(SiO₂), later, and

2 retained clock output for a second step, or approximately 24 krad(SiO₂), before dropout. Seven devices retained a valid clock signal through at least one further dose step, 11 to 24 krad(SiO₂) (median 14.5) past their digital failure, with no clock failure ever observed. The remaining 7 were not irradiated past their first point of failure. A minority of devices demonstrated recovery from digital failure, with 4 second-campaign and 5 first-campaign devices resuming valid communication after their initial fault. Recovery was observed only while the clock remained operational, and no statically biased device recovered once the clock had failed, suggesting that digital failure is a potentially recoverable event and clock failure is terminal unless the device is deliberately annealed. No strong mode-dependent trends were observed regarding these failure patterns.

IV. PREDICTIVE FAILURE ANALYSIS

Because of the large variability in failure points for this system-on-chip (SoC), the typical worst-case qualification of roughly 60 krad(SiO₂) used by mission designers leaves a roughly 120 krad(SiO₂) spread of observed failures unaccounted. A supplemental screening method utilizing specific application parameters, such as the target programmable bias modes, or LPMs, could be used to better inform these designers about what a realistic mission lifetime could be. As results demonstrated large variability even within bias modes, however, a maximally informative model must further be informed by some feature(s) related to the innate silicon variability that causes these discrepancies in failure.

The internal VLO is a native resistor-capacitor (RC) relaxation oscillator that is known to be highly variable between devices, with baseline frequencies ranging from 6 to 14 kHz [10]. Unlike the DCO in the DUT, which is factory-trimmed to hold a calibrated target frequency, the VLO runs uncompensated and untrimmed, so its output more directly reflects the state of the underlying silicon. This clock is used for extremely low-power applications where precise timing is not essential, such as a periodic wakeup timer from a deep LPM. In this study, the VLO is analyzed to determine if its innate variability correlates with the same variability that causes discrepancies in failure points between devices.

The primary device-specific features used throughout this work are statistics of the captured VLO clock signal, and they divide into two physically distinct families, illustrated

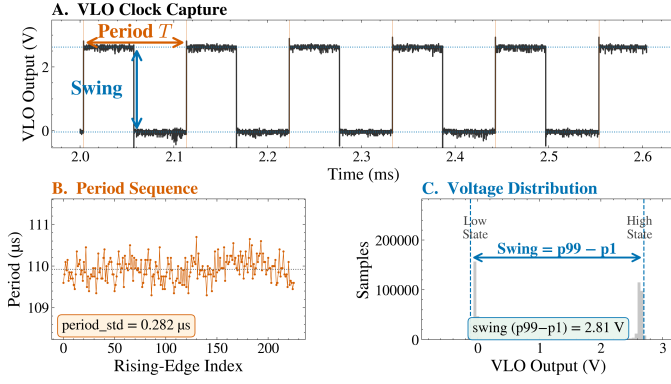


Fig. 3. Extraction of the predictive features from a single raw VLO capture. (a) One capture window, marking the rising-edge period T and the rail-to-rail voltage swing. (b) The per-cycle period sequence, whose spread defines the jitter statistics. (c) The voltage histogram, whose low and high populations define the swing and per-rail noise.

in Fig. 3. Amplitude features describe the voltage distribution once the waveform’s low and high states are separated by an Otsu threshold, a standard technique that places the boundary where the two sample populations are best separated [11]. The rail-to-rail swing, the span between the 99th and 1st voltage percentiles, captures the output drive strength, while `raw_low_std` and `raw_high_std`, the standard deviations of the two separated states, capture the noise on each rail.

Period features instead describe the sequence of rising-edge crossing times. The mean period (`period_mean`) reflects the oscillation frequency set by the RC time constant. The standard deviation (`period_std`) and the coefficient of variation (`period_cv`) measure cycle-to-cycle jitter, with the latter normalizing the jitter by dividing the standard deviation by the average period. The first-to-last-cycle change (`period_delta_last_first`) measures any drift across the capture window. The intrinsic manufacturing variability affecting transistor thresholds and the oscillator’s timing network are revealed in these features. Additional metrics of each clock capture were calculated and deployed during modeling and are discussed where they survived selection methods.

A. Evaluation Protocol

For a supplemental pre-screening method, only the pre-irradiation features of the oscillator can inform the model. The targets for prediction, or labels, are ideally the exact point of device failure. However, the failure dose retained from these experimental campaigns is known only as the interval described in Section II-B, so the label for each device must instead be the midpoint of its interval. Devices that survived the test limit, shown in Fig. 2 as right-censored, are retained by the survival models and scored one-sided, with no penalty for a prediction above the last tested dose.

Two metrics summarize model performance. The first is Harrell’s concordance index C [12], which measures how well a model orders devices by failure dose. Every pair of devices whose order is known contributes, which requires at least one observed failure in the pair. A pair is concordant when the device assigned the higher risk, equivalently the

lower predicted dose, is the one that fails first. The index is the fraction of orderable pairs that are concordant, with ties counted as one half. A value of 0.5 is no better than chance and 1.0 is a perfect ordering. Concordance does not penalize a prediction that is shifted uniformly across every device, since only the relative order matters. As the current dataset consists of only two lots, a model that omits the lot label carries a systematic offset influenced by the between-lot mean difference, shrinking as the model recovers that separation from other features. That offset inflates an error metric but leaves the device ordering intact, so concordance measures the signal such a model can realistically recover. Its statistical power scales with the spread of the doses being ranked, making it most informative across the full failure range.

The second metric is the interval-aware mean absolute error (MAE). Because the true failure dose is known only to lie within a device’s window, a prediction landing anywhere inside that window is scored as zero error, and one outside is charged the distance to the nearer edge. The reported MAE is the mean of this per-device interval error. Coverage, the fraction of predictions landing inside the window, is reported beside it as a diagnostic and is not considered for rankings. It confirms that a low MAE reflects predictions genuinely inside the windows rather than a favorable average over near misses.

The pre-irradiation models were evaluated over fifty train/test splits at an 80/20 ratio, stratified within each (lot, mode) cell so that every cell contributed its own 20% to each test set rather than the split being drawn from the pooled device list. This stratification was necessary because the cells contain only 4 to 12 devices, and an unconstrained shuffle regularly produces test sets in which the smallest cells are absent entirely, leaving those modes unevaluated for that split.

Repeated splits of one fixed cohort share most of their training devices, so the fifty per-split scores are far from independent, and a naive confidence interval over them shrinks with the number of splits even though no new information is added. All bracketed intervals in this paper are 95% confidence intervals of one of two forms, and which form applies is set by what the statistic averages over. A quantity that is itself a mean across the fifty repeated splits uses the Nadeau–Bengio correction [13], which inflates the between-split variance to account for the training devices the splits share. For this cohort, a naive interval over the splits would otherwise be up to roughly $4\times$ too narrow. The other intervals apply to statistics computed over a fixed pool of predictions, many of them from the same device. A device-cluster bootstrap resamples whole devices with replacement, carrying all of a device’s predictions together rather than resampling them one at a time. This matches how the data were gathered and prevents a device’s correlated predictions from being treated as independent.

B. Pooled Prediction and Naive Baselines

A single model trained across all ninety-five devices establishes how much the oscillator adds beyond what the bias mode and lot already reveal. The model is a gradient-boosted

survival ensemble. Rather than predict a single failure dose, it estimates each device's probability of failure as a function of dose. That framing is what lets it train on the right-censored survivors. A device that reached the test limit without failing constrains its failure probability to stay low through its last tested dose, so the model folds it in as a lower bound rather than discarding it or assigning an invented failure dose. The ensemble also captures nonlinear interactions among the period features without manual specification. It receives the bias mode as a one-hot encoding and the VLO period-jitter statistics, and it is scored against two featureless cell-mean lookups, a per-mode lookup and a per-(lot, mode) lookup.

Results from these models, as well as the spread of a key feature on which they were trained, are presented in Fig. 4. The pooled VLO model, given no lot label, reaches a concordance of 0.81 and a mean interval error of 4.4 krad(SiO₂). It cuts the bias-only error by roughly two thirds and recovers much of a lot-aware lookup's accuracy without ever being told the lot, using the period statistics to stand in for the missing label. A per-mode cell-mean lookup, knowing only the bias mode, orders the devices at a concordance of 0.71 and an MAE of 12.6 krad(SiO₂), while a per-(lot, mode) lookup that is handed the lot label reaches a concordance of 0.80 and an MAE of 2.4 krad(SiO₂), showing that lot and mode in these datasets carry most of the explainable variance. Supplying the lot to the VLO model leaves its pooled error essentially unchanged at 4.1 krad(SiO₂), so the oscillator's primary value here lies in reconstructing the lot.

The same model trained on the factory-trimmed DCO, captured on the same pin with identical feature extraction, does not reproduce this result. Its concordance of 0.76 and its mean interval error of 12.9 krad(SiO₂) overlap the bias-mode-only baseline, so the calibrated oscillator supplies little to no usable failure information. The cause of this difference in performance between VLO and DCO features can be better understood by analyzing the spread of the `period_cv` feature between the two oscillators, presented in Fig. 4(b). Across devices, the VLO period jitter in the form of `period_cv` spans an interquartile range near 38% of its median; its absolute jitter, `period_std`, correlates with the failure dose (Spearman 0.24, $p = 0.02$), while the DCO jitter is regulated to roughly 1% and is uncorrelated. The factory calibration of the DCO removes the part-to-part variation leveraged here that the VLO preserves.

C. Per-Mode Refinement and the Lot Premium

The pooled model mixes all six bias modes, so it cannot reveal whether the oscillator carries failure information within any single mode. Fitting one model per bias mode isolates each mode's contribution and lets each mode select the feature family and regressor best matched to its degradation. Within a single mode, the failure doses occupy a much narrower band than the full population, so the concordance metric becomes ill conditioned for evaluation. A model can place every device close to its true dose yet still order the mode little better than chance, and with fourteen to eighteen devices per mode, the index carries wide intervals. Each mode's selection is therefore

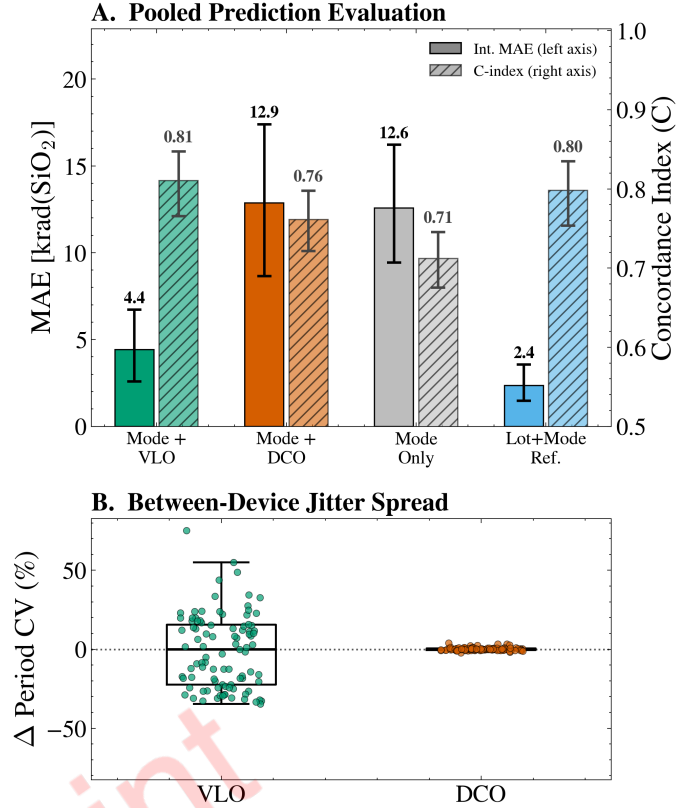


Fig. 4. Pre-irradiation failure-dose prediction using the VLO or the DCO features. (a) Concordance index (hatched bars, right axis) and interval-aware mean absolute error (solid bars, left axis) for two lot-blind gradient-boosted survival models, *Mode + VLO* and *Mode + DCO*, and two cell-mean lookups. The *Mode Only* lookup assigns each device its bias-mode cell mean. The per-(lot, mode) *Lot+Mode Ref.* lookup is handed the lot label directly. (b) Between-device spread of the period coefficient of variation (CV), defined as the standard deviation of the period sequence normalized by its mean, plotted as the percent deviation from each clock's population median. Each point is one device. Error bars are 95% device-cluster bootstrap confidence intervals.

judged by the interval-aware MAE against the best featureless constant for its cell.

Each of the six bias modes is assigned its own model type and feature set, selected from a grid of eight model types and twenty-eight feature sets. The model grid spans five families chosen for complementary strengths. Penalized linear regressors (Lasso, ridge, and elastic net) test whether a simple weighted sum of features suffices and yield interpretable coefficients. Random forests capture nonlinear interactions among features, with tree depth capped to suit the small cells. Accelerated-failure-time survival fits incorporate the right-censored survivors directly through their likelihood, a censored device contributing a lower bound rather than being dropped. Gradient-boosted survival ensembles, the family used by the pooled model of the previous section, add boosted nonlinear survival fits that likewise use the right-censored survivors. Quantile regressors at the median and the 90th percentile target the interval directly rather than the conditional mean. Featureless per-cell constants, the training mean and the training median of each (lot, mode) cell, compete in the same grid, so a learned model is reported only where it beats a lookup table.

With the large grid of model types and feature sets used

competing for as few as fourteen devices in a mode, some configuration will win any single split by chance alone. Choosing each mode's configuration on the same fifty splits used to score it would let that chance winner report its own optimistically low error. The configuration is therefore chosen by a leave-one-split-out procedure. Each of the fifty splits is held out in turn. The candidate configurations are ranked by their mean interval-aware MAE on the other forty-nine splits, and only the top-ranked configuration predicts the held-out split. Because every reported prediction comes from a configuration selected without ever seeing the split it is scored on, the act of selection cannot inflate the result. The 90th-percentile interval error accompanies each selection in TABLE II as a worst-case tail statistic and does not enter the ranking.

The per-mode selections settle on a few compact feature groups drawn from this pool. A ten-feature extended predictive set anchors the deep-mode selections. It grows from a two-feature amplitude core, `robust_cv` and `iqr`, retained from early screening as the pair whose within-cell correlations reproduced most consistently across lots and acquisition repeats. A Spearman correlation scan across per-(lot, bias) device subgroups then adds any feature that correlated with failure dose at $p < 0.05$ in at least one subgroup: the amplitude statistics `rms`, `p99_minus_p1`, `p95_minus_p5`, and `entropy`, and the period statistics `period_std`, `period_iqr`, `period_p95_minus_p5`, and `period_masd`. A second, purely period-based group collects four period-spread statistics, `period_std`, `period_mad`, `period_iqr`, and `period_p95_minus_p5`, into a single mechanistic jitter family.

Two further families build on the wavelet-scattering transform (WST) [14], a cascade of wavelet convolutions and modulus nonlinearities followed by time averaging. The construction yields a translation-invariant, multi-scale description of a signal that is stable to small deformations, capturing structure at and across time scales that single summary statistics reduce to one number each. A third feature family applies the WST to the raw voltage waveform, producing a 374 coefficient expansion that is distilled onto the K supervised partial-least-squares directions most covariant with the training labels, refit inside every training fold so that no test label can influence the projection; these are the PLS- K sets, and PLS-1 can also pair with the extended set as a hybrid. A fourth applies the WST instead to the sequence of 128 successive rising-edge periods, so its 84 coefficients summarize jitter and jitter-of-jitter across time scales with no amplitude content at all. The period-spread and period-WST families are the only ones admitted to the lot-blind selections, for the reasons Section IV-D details. Additional families, including the catch22 canonical time-series statistics [15] and a pretrained tabular foundation model [16], were evaluated under the same protocol and did not displace any per-mode selection.

TABLE II lists the selection, error, and constant floor for every mode under both deployment scenarios. With the lot known, the per-mode models reach a mean interval error of 2.2 krad(SiO₂). Pooling their predictions into a single ranking gives a concordance of 0.843, reported for comparison with the pooled model of the previous section, although the per-mode

configurations themselves are selected on the interval-aware MAE. The oscillator features improve upon the featureless constant in LPM0 (the scattering-PLS set, the constant carrying 19% excess error over the model) and in LPM2 and LPM3 (the extended set, 142% and 59%). AM, LPM1, and LPM4, however, reduce to a cell constant's prediction.

The improved error predictions relative to the cell constant are tested with a label-permutation null. Within each (lot, mode) cell, the failure-dose labels are shuffled across devices. On every shuffle, the entire grid of model types and feature sets is selected anew on the shuffled labels, and the best candidate found is scored against the best refit cell constant. The p -value is the fraction of 100 shuffles whose best-candidate margin meets or exceeds the real one, so the optimism of having chosen the best of many candidates is priced into the null itself. Chance alone hands the best of this grid a margin of 0.3 to 0.6 krad(SiO₂) over the constant in cells this small, and no mode clears the test (LPM0 $p = 0.33$, LPM2 $p = 0.15$, LPM3 $p = 0.33$). A weaker null that holds the winning configuration fixed and refits only it is cleared by these three modes ($p = 0.025$, 0.010, and 0.035), so the selected configurations track structure beyond a random relabeling once chosen. The margins themselves, however, do not separate from the act of selection at fourteen to eighteen devices per mode. The with-lot improvements are therefore reported as point estimates rather than demonstrated device-level oscillator signal, with AM, LPM1, and LPM4 falling back to population constants outright. The extended feature set's role in the deep modes is visible at the single-feature level: two of its amplitude members carry the strongest within-lot associations with failure dose, `entropy` at Spearman 0.51 and `robust_cv` at -0.45 , consistent in both lots, where every period-derived feature stays near 0.3 or below.

A lot-blind variant of this selection was also deployed to investigate the value of these oscillator features as both a within-mode and a between-lot discriminator. A lot-blind model is free to reconstruct lot identity from its inputs, which makes this scenario a lot-level claim, so the candidate pool is restricted to the period-derived families, the four spread scalars and the period-WST coefficients, that survive the campaign-confound scrutiny described in Section IV-D. The lot-blind results are tested with the same selection-aware null as the with-lot grid, shuffling within each mode and repeating the entire confound-safe selection on every shuffle.

The results for the lot-blind models are presented in the lower half of TABLE II and visually in Fig. 5, showing the per-device predicted against true error. Without lot as a feature, the pooled per-mode error degrades from 2.2 to 4.8 krad(SiO₂), which is still far below the 12.6 krad(SiO₂) a deployment with no lot records and no oscillator would face. This lot premium concentrates exactly where the lot gaps of Fig. 2 are largest, 400% in LPM0, while AM holds its premium to 26% because its selection, a survival ensemble on the period-WST coefficients, rebuilds the missing label from jitter structure alone. This feature was attempted with all bias mode models, but only showed this prominence in AM. Both modes clear this selection-aware permutation null (AM with $p = 0.005$ and LPM0 with 0.010), decisively carrying value in lot delineation

TABLE II
PER-MODE SELECTIONS FOR THE LOT-AWARE AND LOT-BLIND SCENARIOS.

Mode	AM	LPM0	LPM1	LPM2	LPM3	LPM4
<i>With lot information (pooled $C = 0.843$, $MAE = 2.20$ krad(SiO_2))</i>						
Constant MAE [krad(SiO_2)]	4.22	3.37	2.55	1.69	1.26	1.22
Selected model	cell median	LassoCV	cell mean	LassoCV	RF	cell mean
Feature set	(constant)	PLS-5	(constant)	ext. pred.	ext. pred.	(constant)
Model MAE [krad(SiO_2)] ^a	4.22	2.83	2.55	0.70	0.79	1.22
Model p_{90} [krad(SiO_2)] ^f	6.80	7.71	6.81	2.45	4.56	3.10
Model premium ^b	—	+19%	—	+142%	+59%	—
Coverage	0.68	0.34	0.40	0.69	0.73	0.59
Permutation p ^{c,d}	n/a	0.33	n/a	0.15	0.33	n/a
<i>Lot-blind deployment, period-derived families only (pooled $C = 0.829$, $MAE = 4.77$ krad(SiO_2))</i>						
Selected model	GBS	RF	RF	ElasticNetCV	RF	QuantReg
Feature set	period WST	period spread	period spread	period spread	period WST	period WST
Model MAE [krad(SiO_2)]	5.33	14.16	2.63	1.49	0.89	0.78
Model p_{90} [krad(SiO_2)] ^f	7.14	52.4	6.17	5.23	3.70	2.95
Lot premium ^e	+26%	+400%	+3%	+113%	+12%	−36%
Coverage	0.61	0.28	0.39	0.48	0.67	0.75
Permutation p (lot-blind) ^{c,d}	0.005*	0.010*	0.32	0.47	0.22	0.24

^a Bold marks the selected model's MAE below its per-cell constant. ^b (constant MAE − model MAE)/model MAE, the excess error of the featureless constant. ^c Selection-aware label shuffles: the entire selection is repeated inside each shuffle and its best candidate is scored against the best refit constant. With-lot tests shuffle within each (lot, mode) cell over the full grid (100 shuffles); lot-blind tests shuffle within each mode over the confound-safe menu (200 shuffles). ^d * marks $p < 0.05$. ^e $(MAE_{lot-blind} - MAE_{with-lot})/MAE_{with-lot}$. ^f 90th-percentile interval error over all pooled test predictions.

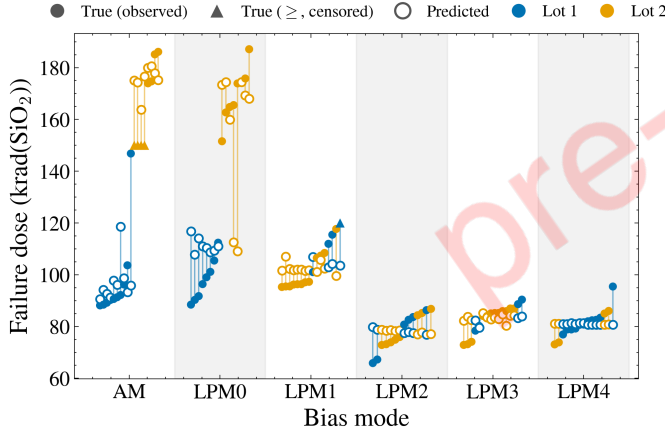


Fig. 5. Lot-blind per-device predictions against observed failure dose, one bias mode per column, colored by lot (lot identity is not given to the model; color is a visualization aid only). Filled circles are observed failure doses. Upward triangles mark right-censored devices, whose failure dose is known only to be at or above the plotted value. Open circles are the leave-one-split-out predictions from each mode's selection-aware lot-blind pick, restricted to the confound-safe period-derived feature pool. Connecting lines are dashed and faded where a censored device's prediction already clears its censoring bound, since no error is charged in that case.

but not necessarily in recovering within-lot variation between devices. LPM0's predictions reproduce roughly half of its separation. No other mode clears the permutation null (LPM1 $p = 0.32$, LPM2 $p = 0.47$, LPM3 $p = 0.22$, LPM4 $p = 0.24$). The deep modes' premiums are modest fractions of small errors, 113% in LPM2 and 12% in LPM3 against errors of 1.49 and 0.89 krad(SiO_2). LPM4's error improves without the lot label, a premium of −36%, consistent with its two lots agreeing in failure dose within 2 krad(SiO_2). Splitting that mode by lot costs sample size without adding separation.

These methods are not able to fully produce the MAE that direct lot encoding produces, but they do preserve the ordering. The pooled concordance of all lot-blind, per-mode models reaches 0.829, slightly below the lot-informed per-mode variants at 0.843, and above both the pooled Mode + VLO model and the Lot + Mode lookup (0.81 and 0.80), showing that while lot as a feature dominates the MAE, VLO features encode the device failure ordering. Because of this, VLO features show use in screening practices not necessarily as a direct failure dosage indicator, but in supporting the ordering of the most resilient configurations and devices.

D. Lot Discrimination and the Campaign Confound

Lot-level observations are confounded by separate testing campaigns in this work. The irradiation chamber, hardware, firmware, and ambient conditions across both campaigns were consistent, limiting potential confounds, but not completely eliminating this uncertainty. Testing every pre-irradiation feature of both clocks directly as a lot discriminator shows one nuance between campaigns concretely. Classifiers on the amplitude families appear to separate the lots well, but the amplitude statistics quantize at the oscilloscope's roughly 40 mV vertical resolution and the two campaigns occupy different discrete levels. This is most visible in the DCO's amplitude interquartile range, which takes one of four levels between 2.29 and 2.41 V on every first-campaign device but concentrates at a single 2.49 V level on 41 of the 52 second-campaign devices. That is likely an instrument artifact, not silicon variability between lots, so amplitude features are excluded from all lot-level claims for both clocks.

The period family survives the same scrutiny. The strongest VLO period statistics separate the lots with standardized effect

sizes (Cohen's d) as large as 2.3, while the DCO's period features, captured minutes apart on the same pin, reach a Cohen's d of at most 0.4, far below the VLO and consistent with no lot separation. A session artifact would be expected to appear in both clocks, so the dissociation instead indicates device-resident process variation that the DCO's factory trim calibrates away. Conclusive attribution still requires testing multiple lots within a single campaign, which is left to future work. The within-lot device-level results of this paper are unaffected by the confound, since the comparisons operate within (lot, mode) cells where the campaign is constant, and the within-cell use of amplitude statistics is likewise safe because every device in a cell shares one instrument configuration.

The same exclusion sets the lot-blind candidate pool of TABLE II, which admits only the period-derived families: the four period-spread scalars and the period-WST coefficients. Both are computed from rising-edge times alone, and the period-WST coefficients show none of the discrete instrument levels that flag the amplitude families. Unrestricted selection was attempted and showed better lot-blind configurations in three modes. A scattering-PLS hybrid reaches 10.1 krad(SiO₂) in LPM0 against the period families' 14.2, and extended-set and hybrid picks reach 0.66 and 0.71 krad(SiO₂) in LPM2 and LPM3 against 1.49 and 0.89. Each of those configurations leans on amplitude-domain or raw-waveform features, so a lot-blind model built on them could recover lot identity from the instrument signature rather than the silicon. Those figures are reported here only as the apparent ceiling of the confounded families and are kept out of TABLE II and every lot-level claim.

V. IN-SITU HEALTH MONITORING

As the authors in [8] showed with a convolutional neural network, the VLO exhibits features that allow binary classification of a device into a dosed (greater than 25 krad(SiO₂)) or un-dosed category. These observations are leveraged here to form a more application-focused predictive model in which the remaining dose a device can absorb before failure is estimated, raising a direct warning of imminent failure as that margin lessens. Each in-situ capture serves as a prediction point, and the per-capture VLO features described above, together with their deltas from the device's own pre-irradiation baseline, serve as the condition indicators. The campaign-level amplitude effects discussed in Section IV-D are handled differently here. The baseline-referenced delta features are campaign-immune by construction while raw swing retains a modest campaign association. Therefore, only the baseline-referenced swing delta is used to eliminate confound risk.

The evolution of these indicators with TID has been previously observed and characterized. Charge trapping shifts threshold voltages and increases leakage, which manifests first as frequency drift and degraded edge-timing stability [9]. Degraded timing margins are an established precursor of functional TID failure in microcontrollers [17], and oscillator frequency degradation is an established TID observable in dedicated ring-oscillator test structures [18]. As damage accumulates further, the output drive weakens and the rail-to-rail

amplitude of the waveform collapses, an end-stage form of the output-swing reduction and edge slowing expected from TID degradation of CMOS output buffers [19], [20]. The period family is therefore expected to track accumulated damage through mid-life, with its location statistics drifting steadily and its jitter rising as failure nears, while the amplitude family is expected to activate near the end of life. Captures of the observed failure dose are excluded from all training and evaluation, as that point coincides with the failure itself and carries no actionable lead time; every earlier capture, including the last good one before failure, is retained.

A. Remaining-Dose Estimation

Seven feature configurations were used to evaluate remaining-dose regression, consisting of dose only, dose with bias, bias with VLO period features, VLO period features only, dose with bias and VLO period features, dose with bias and the VLO's rail-to-rail swing, and dose with bias and VLO period features augmented with its rail-to-rail swing. Each was evaluated over fifty device-level splits. Each split assigns whole devices to training or test at a 70/30 ratio. The ratio is 70/30 rather than the 80/20 used for the pre-irradiation models because the unit of evaluation is the capture, and thirty percent of devices supply roughly 230 held-out captures per split. This yields enough information to resolve the per-band metrics described below, while the pre-irradiation cohort's cells of 4 to 12 devices cannot spare more than twenty percent of each cell without starving training. These splits are not stratified by lot or mode, because the in-situ model is a single pooled regressor scored by remaining-dose band rather than per cell, and a device-level shuffle leaves every band well populated in each split. Within each feature configuration, the regressor was chosen from six possible options, including gradient boosting under absolute-error and quantile losses, random forest, LASSO, ridge, and median quantile regression. These choices were made by the same leave-one-split-out procedure as the per-mode models, ranking candidates on the other forty-nine splits. Gradient boosting was selected in every configuration of TABLE III across all fifty folds, so the table isolates the effect of features rather than of model choice.

Each model predicts one continuous quantity, the dose a device can still absorb before failure, and is trained on every retained capture from every device alike. The bands enter only at evaluation, grouping each capture's error by its proximity to that device's own failure. The close band is the two retained captures nearest failure, the regime in which failure is imminent. It is defined by capture rank rather than a dose cutoff to accommodate the variation in dose step between devices, which ranged from 11.2 to 17.3 krad(SiO₂). The mid band holds the remaining captures within 60 krad(SiO₂) of the failure midpoint, where degradation is well underway, and the far band holds those beyond it, early in life where the least degradation has occurred. Every device contributes to all three bands, so each reflects the whole cohort rather than a long-lived subset. The far band thins only at its earliest extreme, where the highest-remaining captures belong to the longest-lived devices alone.

TABLE III
IN-SITU REMAINING-DOSE PREDICTION ERROR BY FEATURE CONFIGURATION AND EVALUATION BAND.

Dose	Bias	VLO		Far	Mid	Close	1-away	Pooled	Close cov.	1-away cov.	Pooled p_{90}
		Period	Swing								
✓				24.0	5.1	7.6	9.6	14.8	38%	36%	68.2
✓	✓			19.5	3.0	5.5	7.2	11.5	47%	45%	57.5
	✓	✓		20.7	8.8	14.7	16.9	15.8	18%	14%	48.2
		✓		25.6	9.6	16.1	17.7	18.7	14%	10%	53.3
✓	✓	✓		18.3	4.0	6.1	7.6	11.4	44%	39%	48.5
✓	✓		✓	19.3	3.7	5.3	5.9	11.6	53%	61%	53.6
✓	✓	✓	✓	18.4	4.4	6.0	6.5	11.5	51%	55%	47.4

Check marks give the inputs to each model: the external dose record, the programmed bias mode, and the VLO's per-cycle period and rail-to-rail swing statistics. Every configuration uses the same gradient-boosted regressor. Errors are the interval-aware MAE per capture, presented by remaining-dose band and pooled, with the 1-away column restricted to the last good capture before failure (a subset of the close band). Coverage is the fraction of predictions inside the failure window, reported for the close band and at that last capture. The pooled 90th-percentile is the worst-case error over all captures. Errors in krad(SiO₂), coverage in percent. Bold marks the best value per column.

Comparisons between models given access to different features can be seen in TABLE III. The interval-aware mean absolute error per capture is reported by band and at the last good capture before failure, alongside the close-band and last-capture coverage and the pooled 90th-percentile worst-case error, each with 95% device-cluster bootstrap intervals. Results show that adding the VLO period statistics lets the estimate follow the individual device, using its accumulated drift to distinguish a degraded device from a fresh one at the same accumulated dose. Given no dose input at all, the bias-and-VLO model reaches a pooled per-capture error of 15.8 krad(SiO₂), overlapping the confidence interval of the dose-and-bias baseline ([12.1, 19.5] against [7.4, 15.6]), with the period drift standing in for the missing dose record. Added on top of the dose record, the full configuration of VLO period and swing leaves the pooled mean essentially unchanged, 11.4 for the period family and 11.5 for the full configuration. Demonstrable value lies instead in the worst case and in the approach to failure. It cuts the pooled 90th-percentile error from 57.5 to 47.4 krad(SiO₂) and sharpens the estimate as failure nears, where the dose record alone can only fall back on the cohort margin.

The prediction error is not uniform across life, and the same ordering, largest in the far band, smallest in the mid band, elevated again in the close band, appears even in the dose-only row of TABLE III (Far 24.0, Mid 5.1, Close 7.6 krad(SiO₂)), which carries no oscillator feature at all. The pattern therefore belongs to how the bands are defined, not to any one feature family. The bands are carved directly on the true remaining dose, and they differ sharply in width. The far band has no upper bound, capturing any remaining dose of 60 krad(SiO₂) or more, and in this cohort that spans true values from roughly 60 to 186 krad(SiO₂), over four times the spread of the mid band's roughly 27 to 57 krad(SiO₂) window. A far-band prediction is therefore a harder point estimate before any feature is considered, and its error is largest, at 18.4 krad(SiO₂) for the full configuration. The mid band's narrower window carries little directional bias, and its error is smallest, at 4.4 krad(SiO₂). The close band's true-value spread is narrower

still, roughly 3 to 31 krad(SiO₂), yet its error remains above the mid band's. This is the symptom of bias, as most close-band predictions overestimate the device's remaining lifetime. Only devices that lost their clock escape this bias, leaving the digital-first survivors to lift the close-band error to 6.0 krad(SiO₂). Feature richness reduces these tail errors without removing them, cutting the far-band error from the dose-only baseline's 24.0 to 18.4 krad(SiO₂) and the close-band error from 7.6 to 6.0, while the already near-unbiased mid band moves from 5.1 to 4.4. The far-band uncertainty dominates the pooled worst case and is set by the longest-lived devices, almost all of them second-lot parts in the active and lightest-sleep modes.

Among the dose-anchored configurations, the lowest close-band error belongs to dose with bias and the swing alone, at 5.3 krad(SiO₂), with the highest close-band coverage in TABLE III at 53%. The full configuration, which adds the period features, costs a little of this accuracy, 6.0 krad(SiO₂) and 51% coverage, in exchange for the lowest pooled worst-case error. Dose with bias alone reaches 5.5 krad(SiO₂) and 47% coverage, while dose with bias and the period reaches 6.1 krad(SiO₂) and 44%, so the period adds nothing at the close band by itself. On the capture immediately preceding failure, both swing-bearing configurations are the only ones whose coverage improves rather than degrades, dose with bias and the swing rising to 61% and the full configuration to 55%, illustrating their impact as imminence detectors. These margins are tested with the same selection-aware label-permutation null as the pre-irradiation models: on each of 100 within-device shuffles of the remaining-dose targets, the full six-model selection is repeated and the statistic recomputed. The last-capture coverage of both swing configurations exceeds every shuffle (observed 61% and 58% against a null mean near 12%, $p = 0.01$), and the pooled advantage of the dose, bias, and period configuration over dose alone likewise clears its null ($p = 0.01$). The period family's close-band increment does not ($p = 0.60$).

The close-band tail error can be better understood by analyzing the specific failure modes of each device. The first

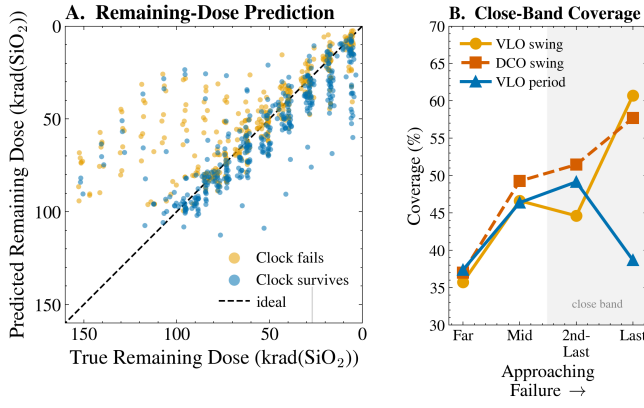


Fig. 6. In-situ remaining-dose estimation. (a) Predicted versus true remaining dose for the full in-situ model, which adds the VLO period and rail-to-rail swing to dose and bias. Each point is one device-capture averaged over the held-out seeds. The dashed line marks ideal agreement. Points are colored by whether the device later loses its clock within the test. (b) Coverage, the fraction of predictions landing inside the failure window, by remaining-dose band advancing toward failure from left to right. Three lines are shown, a dose-and-bias model carrying the VLO swing, the DCO swing, or the VLO period. The close band is split into its last two retained captures before failure.

functional failure of a device does not always include the loss of its clock. Most first failures are a missed reset or interrupt acknowledgment with the clock still running. Grouping the in-situ predictions by clock-failure outcome separates the two cases, as shown in Fig. 6(a). The devices that lose the clock within the test, where the digital fault and clock loss fall within roughly one dose step, carry a close-band interval error of 3.1 krad(SiO₂). The devices whose clock outlives the test carry 9.0 and largely set the close-band tail. The same split holds at the worst case. The clock-failure devices' 90th-percentile error is only 9.9 krad(SiO₂), while the clock-survivors carry 28.7, the swing sharpening the cases whose clock it can sense while over-predicting the digital-first survivors that dominate this tail. The close-band limit is therefore a property of the failure population rather than of the degradation signal, and the oscillator tracks the approaching clock loss far better than a digital fault it never sees.

B. Feature Families and the DCO Control

The period and amplitude feature families, because of their differing degradation profiles, serve prediction distinctly. The period family carries the continuous dose track. Its location statistics, the mean and median period and their baseline drift, drift steadily with dose and hold the remaining-dose estimate across life. Its jitter statistics, the standard deviation, coefficient of variation, and interquartile range, carry the larger share of model weight. The dedicated near-failure signal comes instead from the amplitude family.

Within the amplitude family, the VLO's baseline-referenced rail-to-rail swing delta, the span between the 99th and 1st voltage percentiles, is flat until it collapses near failure, making it an end-of-life signal rather than a continuous tracker. Added on top of the dose-and-bias and period model, the swing raises the close-band coverage, the fraction of predictions landing inside the failure window, from 44% for the period family to

51% (TABLE III). In a dose-and-bias model carrying only the swing, coverage rises from 45% two captures before failure to 61% at the last capture before it, while the period family's falls from 49% to 39% over the same step (Fig. 6(b)). The period family therefore serves dosimetry, drifting with accumulated dose, while the swing serves imminence detection.

The factory-trimmed DCO, captured on the same output pin, indicates that this end-of-life signal belongs to the shared output drive rather than the oscillator core. The same single swing feature, evaluated on the DCO in that same single-feature setting, reaches a close-band error of 5.5 krad(SiO₂) against the VLO swing's 5.3, with comparable coverage (55% against 53%). It sharpens toward failure in step, 51% to 58% on the DCO against 45% to 61% on the VLO. The collapse measured here is strictly the partial reduction before each device's final retained capture. The complete rail-to-rail dropout at the failure capture itself, where the swing falls to a fraction of a volt on both clocks, coincides with functional failure and is excluded from every model, since a signal that arrives only at failure carries no actionable lead time. The usable warning is therefore the partial swing reduction that precedes failure, and it could extend to any observable output clock.

The DCO's period, by contrast, adds nothing to the dose record, holding the pooled error at 12.9 krad(SiO₂) and the 90th-percentile worst case at 55.5, essentially the dose-with-bias values of 11.5 and 57.5, against the VLO period's 11.4 and 48.5. Continuous dose tracking therefore remains specific to the uncompensated VLO. Removing the programmed bias mode from the full configuration costs 2.85 ± 0.91 krad(SiO₂), so knowing the sleep state helps the estimate but is not essential. A single monitor still serves a device of unknown mode at a modest penalty.

The period family's four statistics, `period_mean`, `period_std`, `period_cv`, and `period_delta_last_first`, are all derivable on-chip using local timers sourced with the DCO, which was shown to have essentially no frequency drift with dose. Section II-B describes experimental procedures for these campaigns that deployed these timers during irradiation, outputting the average period of 100 individual VLO periods after each oscilloscope capture over UART. Average frequency computations were observed, during irradiation, to track the frequency calculated from oscilloscope captures closely (Pearson $r = 0.999$, median relative deviation 0.55%, $n = 867$ matched captures across all 97 devices), with agreement holding through life. The standard deviation and the first-to-last drift add no hardware, only a running sum of squares and the first and last timed period alongside the existing running sum, a firmware change rather than a new sensor. A device that already logs its own VLO frequency this way can therefore compute the full period family, and with it the self-dosimetry configurations of TABLE III, using only its own timers. The amplitude family the swing draws from has no equivalent path here. It requires the voltage-domain sampling this study collected with an external oscilloscope, so a self-contained deployment of the swing would need an on-chip comparator or ADC channel on the observing device

rather than a firmware change to an existing timer.

VI. CONCLUSIONS

This work characterizes the large variability of TID-induced failure points in a complex MCU and identifies its sources in programmable sleep states and innate inter-device variability. Manufacturing lot membership is shown to be a major contributor to inter-device variability, with devices that were irradiated in the least restrictive bias states showing the greatest failure-point differences between lots. Additional characterization of inter-device variability leverages the internal VLO of the MSP430 devices tested. Models that are informed by bias, lot, and oscillator characteristics were built both to predict device performance before irradiation and to monitor device health as dose accumulates.

Pre-irradiation, a gradient-boosted survival model that is trained on VLO characteristics and target bias configurations and is never given the fabrication lot orders the failure dose across 95 statically biased devices at a concordance of 0.81 and predicts it to an MAE of 4.4 krad(SiO₂). In comparison, a simple lookup table leveraging bias mode and manufacturing lot information achieves a concordance of 0.80 and an MAE of 2.4, showing the value of the VLO at the whole-population level is in reconstructing lot information. This ability is shown to be unique to the period features of the VLO in the current data, allowing separation of the two fabrication lots with effect sizes as large as 2.3, while the trimmed DCO on the same output pin reaches at most 0.4.

Stratifying model and feature choices by bias mode sharpens the population MAE to 2.2 krad(SiO₂) with the lot known, though the per-mode margins over the bias-informed constants do not survive selection-aware permutation testing at these sample sizes and are reported as point estimates. In AM, LPM1, and LPM4, lot identity in combination with a bias-informed constant is the best available model outright. Further, the lot-blind stratified models are shown to order the failure points of devices at a concordance of 0.829, providing a surrogate screening method for selecting the most tolerant devices.

In-situ, the VLO serves as a dosimetry surrogate, estimating the remaining dose at statistical parity with an external record from the bias mode and the oscillator alone, without a dose input. With a dose record in hand, the period drift adds little near failure, where the dose anchor governs the mean error, but the rail-to-rail swing sharpens the estimate as the margin vanishes, lifting the close-band coverage. Over the dose record, the oscillator's added value is the worst-case error, the reconstructed inter-lot offset, and this near-failure sharpening. The continuous estimate is carried by the period family, whose mean and jitter drift with accumulated dose. The rail-to-rail amplitude swing stays flat until it collapses near failure, supplying a separate end-of-life warning. The factory-trimmed DCO on the same pin isolates the two signals. Its regulated period adds nothing to the dose record, so continuous tracking is specific to the uncompensated VLO. Its amplitude swing, however, collapses alongside the VLO's, locating the imminent-failure warning in the shared output-drive path rather than the oscillator core.

These in-situ configurations correspond to deployment tiers that differ in what a mission can instrument. Where an external dose record already exists, dose with the programmed sleep state is the cheapest accurate estimate, and the VLO adds value only in the worst case and the approach to failure. Where no dose record exists, the VLO alone serves as a self-contained dosimeter, estimating the remaining dose from the oscillator and reconstructing the inter-lot offset with no dose input. Systems that can poll an output clock add the rail-to-rail swing as an end-of-life imminence alarm, which could transfer to any observable clock. The economical tier is therefore set by the instrumentation already available, from a bare dose record, to the self-dosimetry oscillator, to a swing-based imminence warning.

A key limitation of these data and analysis methods is that the prediction target of all models is the first point of functional failure, which was always observed to be a failure to respond to an I/O interrupt. In many devices, the clock failure coincided with this point, but it was not wholly consistent in the entire population. For devices in which clock failure was observed significantly after digital failures, the in-situ models struggled to leverage VLO characteristics to predict failure reliably. In these cases, the model over-predicts the lifetime of the device before its first functional failure, even though these I/O faults were observed to be partially recoverable in some cases. For the pre-irradiation models, however, these discrepancies in failure profiles do not affect performance as strongly, as a device that is ordered as low risk (high survivability) is likely to survive to a higher dose than a high risk device regardless of the relatively minor uncertainty the failure label introduces.

An additional limitation regarding the in-situ models is the potential effect of application temperature variability on leveraged oscillator characteristics. Frequency of the VLO is shown to vary by 0.2% per degree Celsius [10], meaning key features that trained models rely upon could behave erratically in an unpredictable environment. To combat this, more informed models would require irradiation at temperature to determine how exactly leveraged features behave.

Findings demonstrated in this work can be leveraged by both manufacturers and designers looking to qualify and deploy parts in harsh environments. At the qualification stage, a single pre-irradiation oscillator capture is an inexpensive, nondestructive addition to existing screening flows, and is shown to identify the more tolerant individual devices. At the design stage, the failure-free region below roughly 60 krad(SiO₂) observed across both lots, together with the per-mode failure distributions, supports more considerate decisions regarding programming strategies and derating margins, since the choice of sleep state alone shifted typical failure doses by tens of krad(SiO₂). In flight, the same oscillator used in pre-irradiation screening doubles as a health monitor, providing in-situ failure warnings with no dedicated sensor hardware and giving mission teams time to deploy mitigation strategies. These approaches are not necessarily specific to this part. The continuous dose tracking requires an uncompensated internal oscillator, of which the VLO is one example, while the end-of-life amplitude warning proved common to both clocks and may transfer to any device with an observable output clock.

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